

Smart Hostel Managment

5-D Process



BY:
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Problem Statement

Manual handling of hostel complaints (plumbing, electrician, internet, etc.) often leads to skipped issues, no way to track status, and little to no feedback after resolution. This lack of transparency frustrates students, forces repeated follow-ups, and creates inefficiency for hostel authorities, affecting daily operations and overall satisfaction.



Understanding Design Thinking

The Five Stages of Design Thinking



Empathize

Gain a deep understanding of the users and their needs.



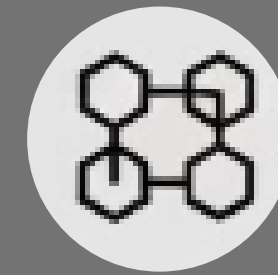
Define

Clearly articulate the problem to be solved.



Ideate

Generate a wide range of creative ideas and solutions.



Prototype

Build tangible representations of the ideas to explore potential solutions.



Test

Evaluate the prototypes with users, gather feedback, and refine the solutions.

Problem Statement

Manual handling of hostel complaints often leads to missed issues, slow updates, and constant follow-ups from students.

Main Idea Thoughts:

Complaints get missed due to no tracking.

Students get frustrated from repeated follow-ups.

What is the problem?

- Complaints for plumbing, electrical, internet, and other issues are managed informally.
- Issues often get skipped or delayed.
- There is no proper system to track what is resolved or pending.
- Students rarely receive updates after submitting a complaint.

How do we know this is a problem?

- Students have to follow up multiple times to get something fixed.
- Some issues remain unresolved for long periods.
- There is confusion about the status of complaints.
- Both students and hostel staff experience frustration and inefficiency.

How will we know when we've solved this problem?

- All complaints are clearly logged and trackable.
- Students receive timely updates without needing to remind anyone.
- Issues get resolved faster and more systematically.
- Both hostel staff and students report smoother operations and higher satisfaction.

Problems with Manual Hostel Complaint Handling

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Impact on Satisfaction and Operations:

Students get frustrated, staff get overwhelmed, and proactive maintenance becomes difficult.

Inefficiency of Manual Systems:

Calls and verbal complaints get messy, things get lost, and staff waste time tracking instead of fixing.

Poor Feedback and No Data Insights:

Feedback and history aren't recorded, so recurring issues can't be identified.

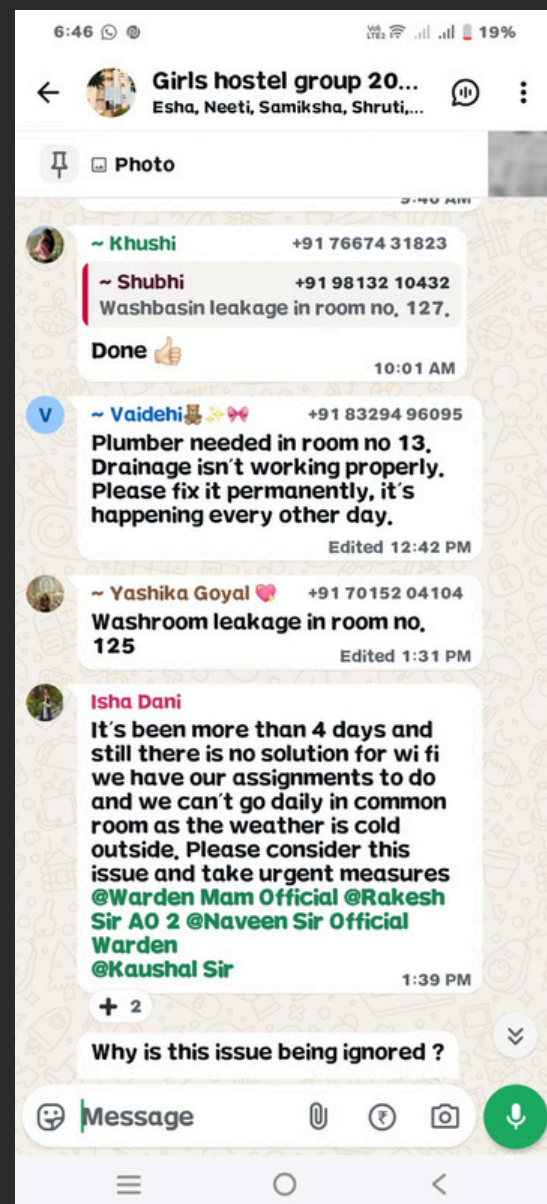
Lack of Transparency and Accountability:

No central system means skipped complaints, no updates for students, and unclear responsibility.

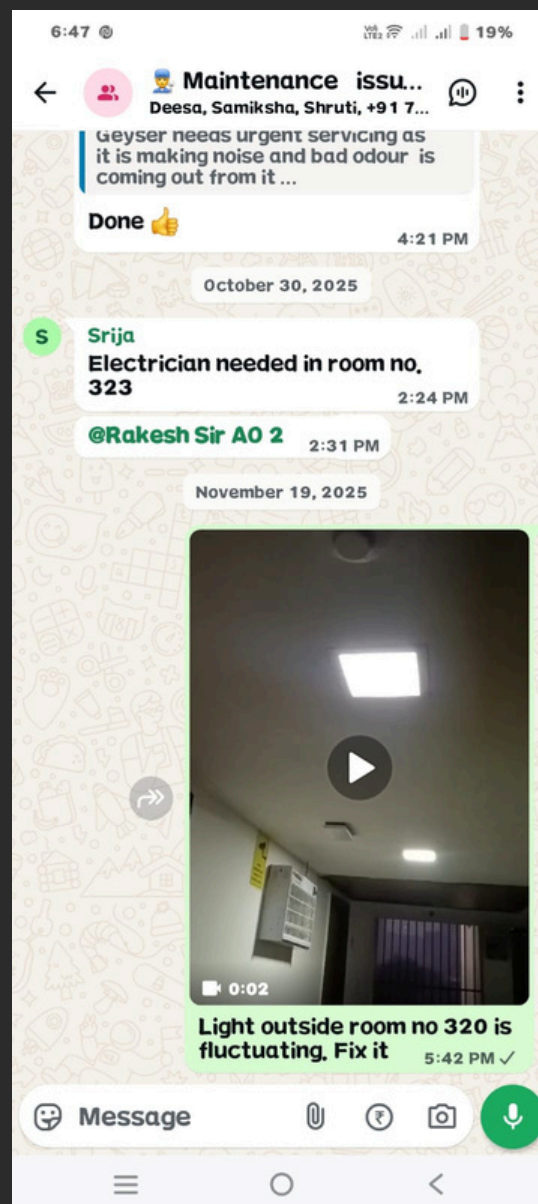
Delayed Resolution and Poor Communication:

Without reminders, responses slow down and students keep chasing updates.

Primary Research



Students using
whatsapp for
complaining



Complaint Register

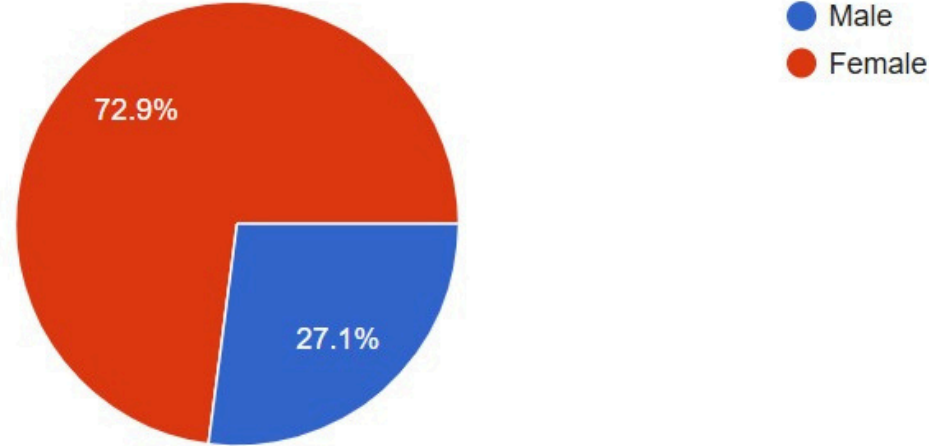
HOSTEL DEFECT COMPLAINT REGISTER					
Date/Month & Time	Room No.	Description of defect	Name and Signature of complainant	Date and Time of Rectification	Signature of maintenance in-charge
24/11/25	4	Washbasin in common room. Please renovate them.	Sarvagata Pandit		
25/11/25	317	Carpenter needed. Lock broken Bathroom.	Monvi	26.11.25	Pranav
25/11/25	317	Electrician needed	Monvi	26/11-25	Komalendra OK
25/11/25	15	Hot spray not working properly	Kapil	26/11/25	Naveen OK
26/11/25	215	Hot spray Broken in washroom & water is leaking	Sakshi Pragna		

Students using pen
and paper for
complaining

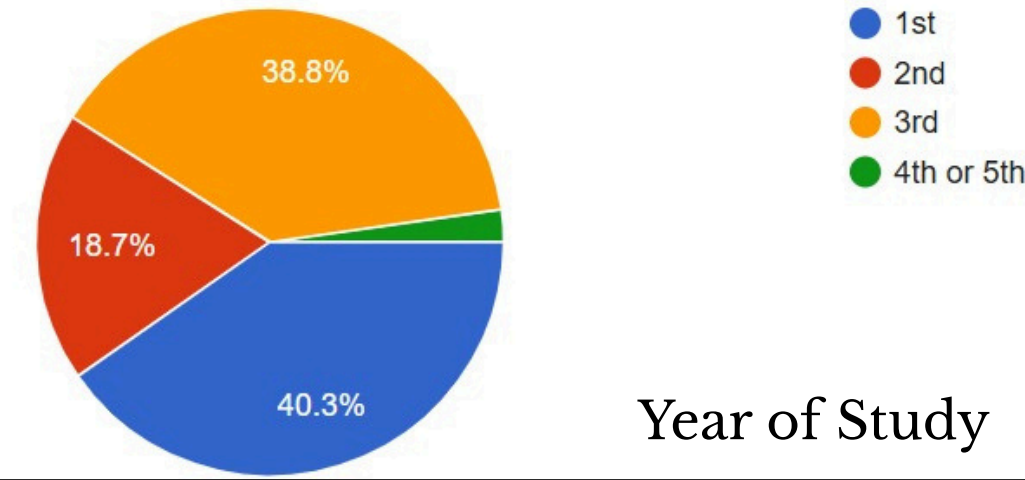
HOSTEL DEFECT COMPLAINT REGISTER					
Date/Month & Time	Room No.	Description of defect	Name and Signature of complainant	Date and Time of Rectification	Signature of maintenance in-charge
8/11/25	111	Wash basin light is not working	Siddhika	10/11/2025	Pooi OK
09/11/2025 09:02 am	324	Carpenter - Almirah HINGE BROKEN.	Vidhi K.	10/11/25	215 OK
09/11/25	320	Carpenter needed, Door lock is broken.	Ayushi	10/11/25	215 OK
09/11/25	218	Electrician needed Washroom light is not working	Payal Rahore	10/11/2025	Pooi OK
09/11/25	108	Electrician needed Geyser is not working	Pooi	10/11/2025	Pooi OK

Insights of Primary research: (Survey Results)

Gender



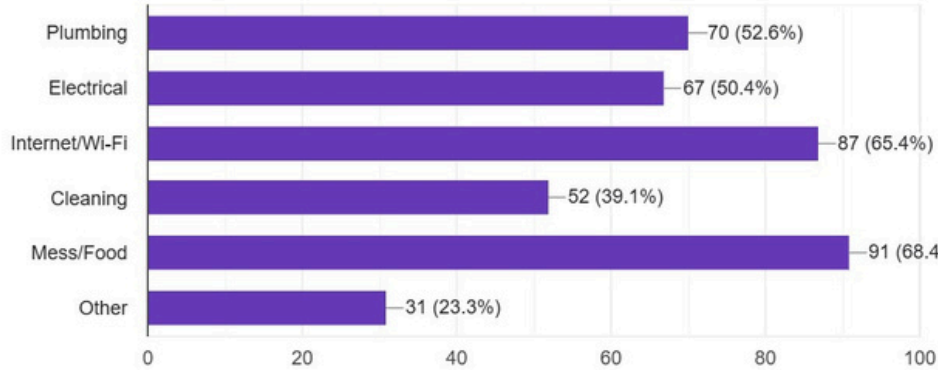
Year of Study



What types of complaints do you usually report? (Select all that apply)

133 responses

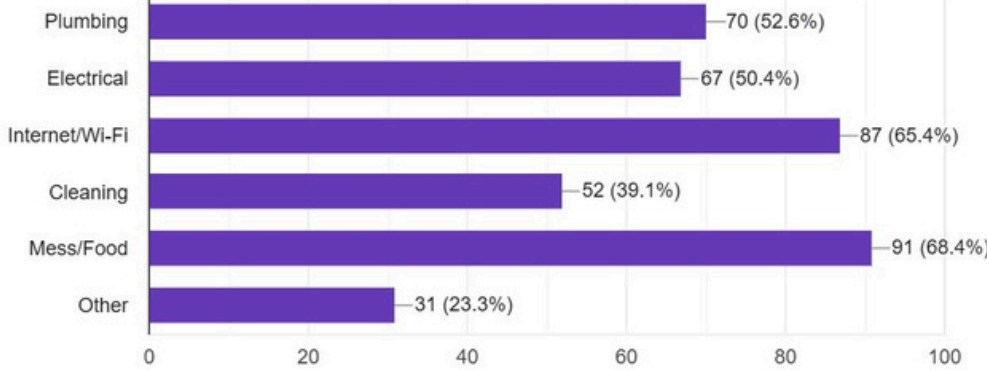
[Copy chart](#)



What types of complaints do you usually report? (Select all that apply)

133 responses

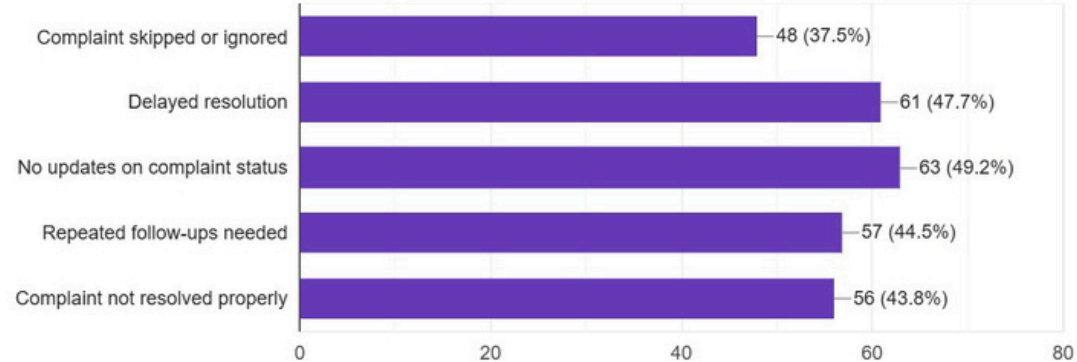
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Have you ever faced these issues with the current system? (Select all that apply)

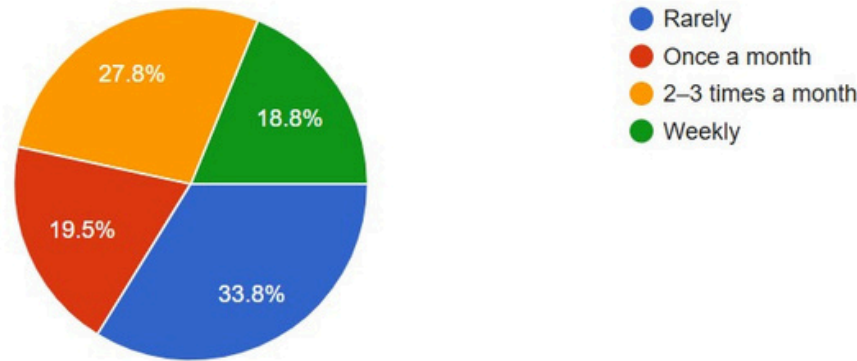
128 responses

[Copy chart](#)



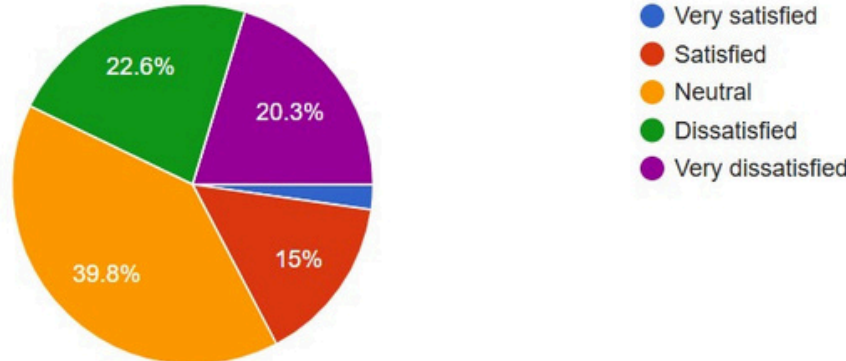
On average, how often do you raise complaints?

133 responses



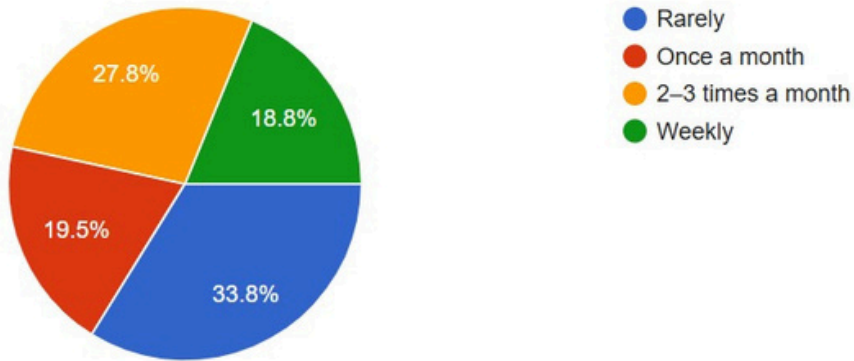
How satisfied are you with the current complaint resolution process?

133 responses



On average, how often do you raise complaints?

133 responses



Interview Questions

Everyday Experiences and Pain Points:

- Can you describe the most recent issue you had in your room?
- What was the problem, and how long did it take to resolve?
- What was the most annoying aspect of the process?

After-Resolution Experience:

- Do you receive any confirmation when your complaint is resolved?
- Are you usually satisfied with the solutions provided?
- Would you like to give feedback after the work is done?
- Have you ever had a problem reoccur because it wasn't fixed properly?

Communication & Reporting:

- At the moment, how do you report a problem by phone, message, or in person?
- Do you find this method convenient? Why not, or why?
- When you file a complaint, do you think it is taken seriously?
- Have you ever had a complaint skipped or ignored? What happened next?

Transparency & Tracking:

- Do you get updates after you report a problem?
- How important is it for you to know the status of your complaint?
- Have you ever been unsure whether your issue will be solved?
- Would you like to see who is assigned to your complaint and when they'll come?



Name : Aditi Sharma
Age : 20 years old
Occupation : Student
Location : KRMU Hostel

User Description:

Aditi is a full-time student juggling academics, club activities, and project work. She spends most of her time in the hostel when not in class or the library. Stable internet, functional electrical appliances, and clean facilities are essential for her studies and well-being.

"I just want a simple way to report problems without running after people. If I can track my food order and bus arrival, why can't I track a complaint in my own hostel?"

Goals:

- Quickly report hostel issues and track their status in real time.
- Get timely resolutions without repeated follow-ups.

Challenges:

- Complaints often get skipped or delayed due to manual handling.
- No transparency or feedback, forcing her to constantly chase hostel authorities.

Needs:

- A centralized platform to submit complaints and receive updates.
- Clear communication and visibility on progress and completion.

Empathy Mapping

SAYS

- “No one is responding.”
- “I don’t know who to contact.”
- “I have to follow up again.”

DOES

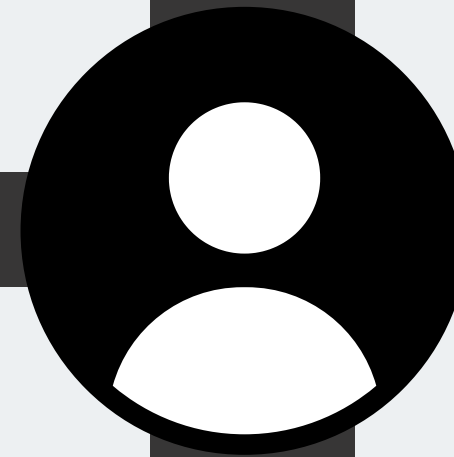
- Reports issues repeatedly.
- Tries to find the warden/technician.
- Waits without knowing the status.

FEELS

- Frustrated.
- Helpless.
- Annoyed by delays.

THINKS

- “My complaint might get ignored.”
- “This system is messy.”
- “There should be a better way.”



5W Questions

What

Hostel complaints often get missed or delayed due to manual handling.

Why

There is no proper tracking or update system in place.

Who

Students and hostel authorities are both affected.

Where

Within hostel premises (plumbing, electrical, internet, etc.).

When

Every time a student raises an issue and waits without clear updates.

403 ERROR

The request could not be satisfied.

Request blocked. We can't connect to the server for this app or website at this time. There might be too much traffic or a configuration error. Try again later, or contact the app or website owner.

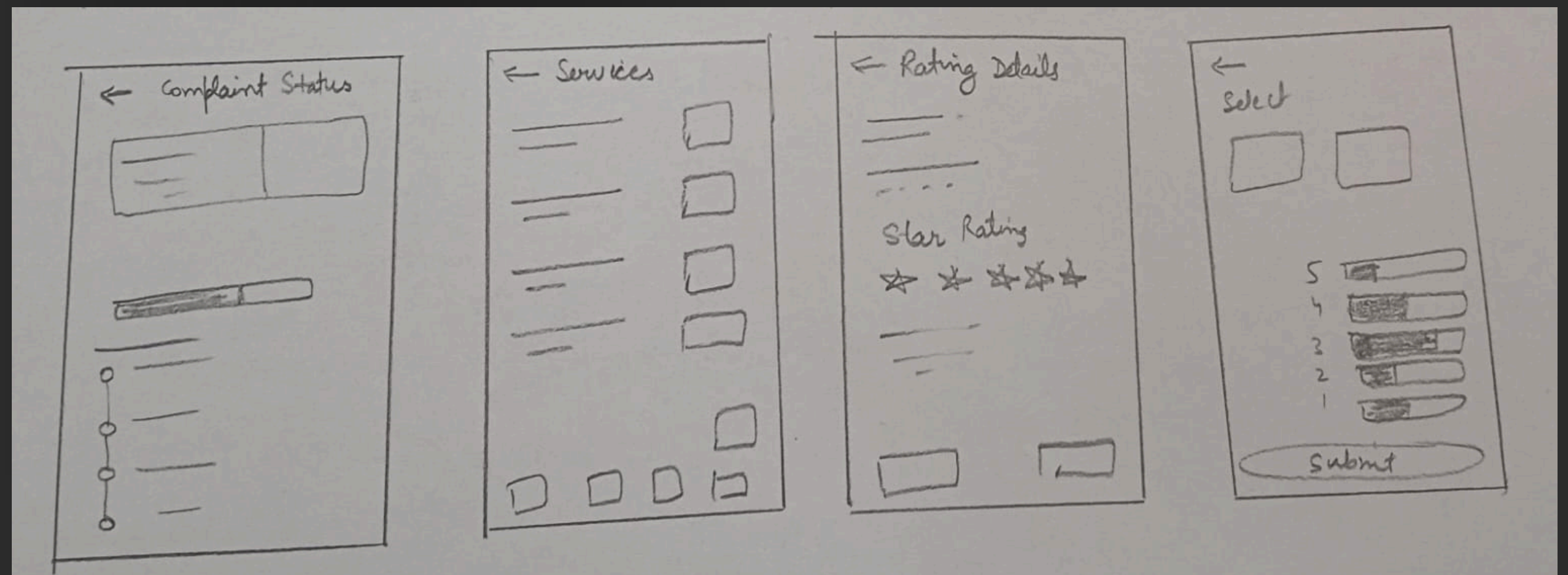
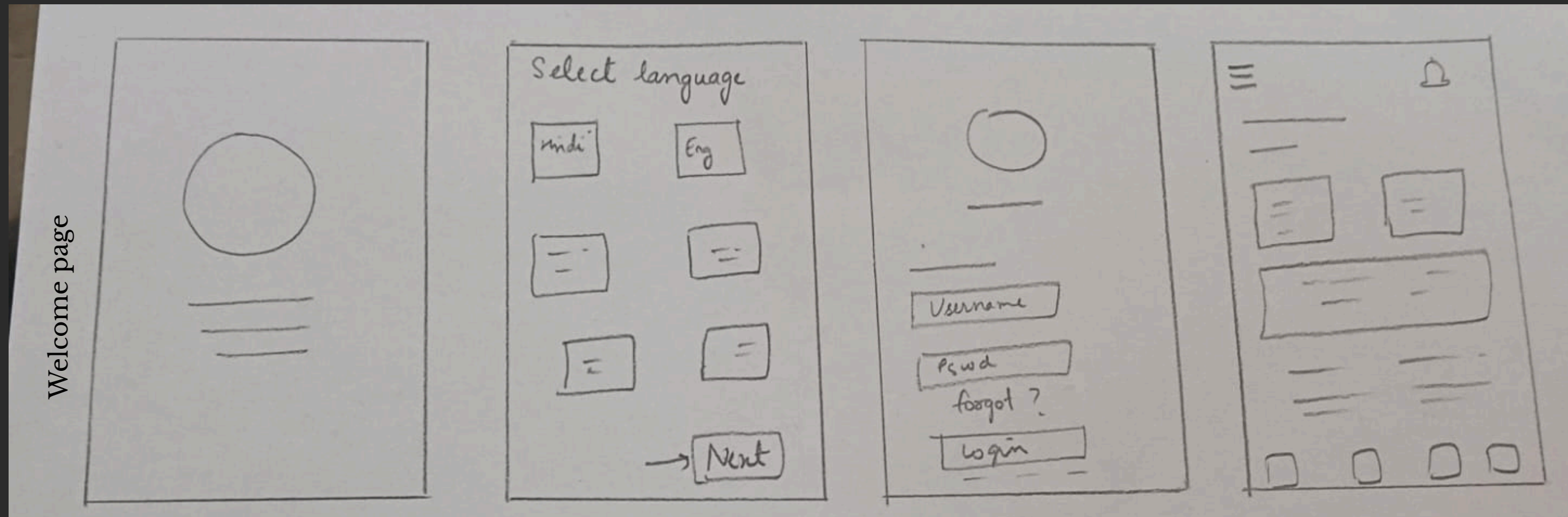
If you provide content to customers through CloudFront, you can find steps to troubleshoot and help prevent this error by reviewing the CloudFront documentation.

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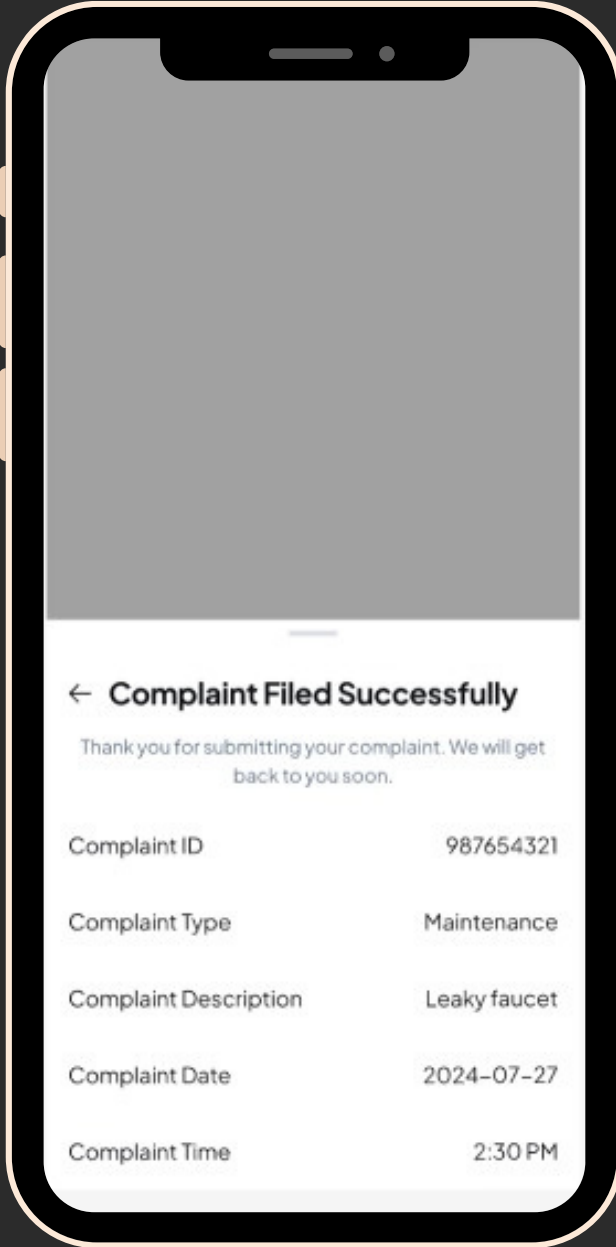
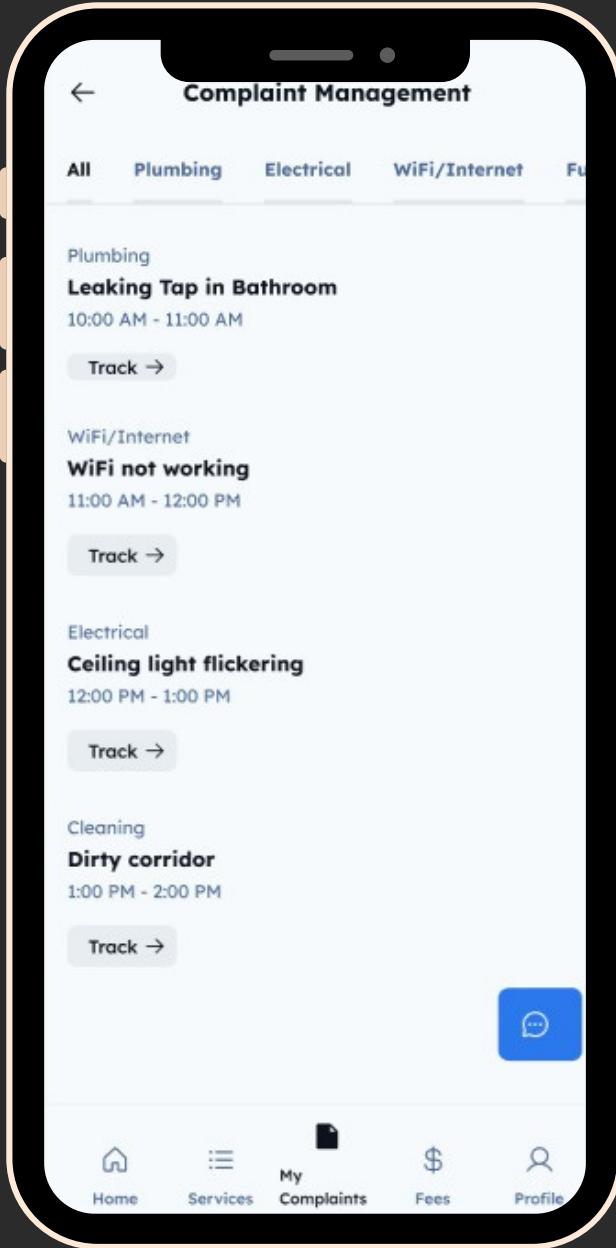
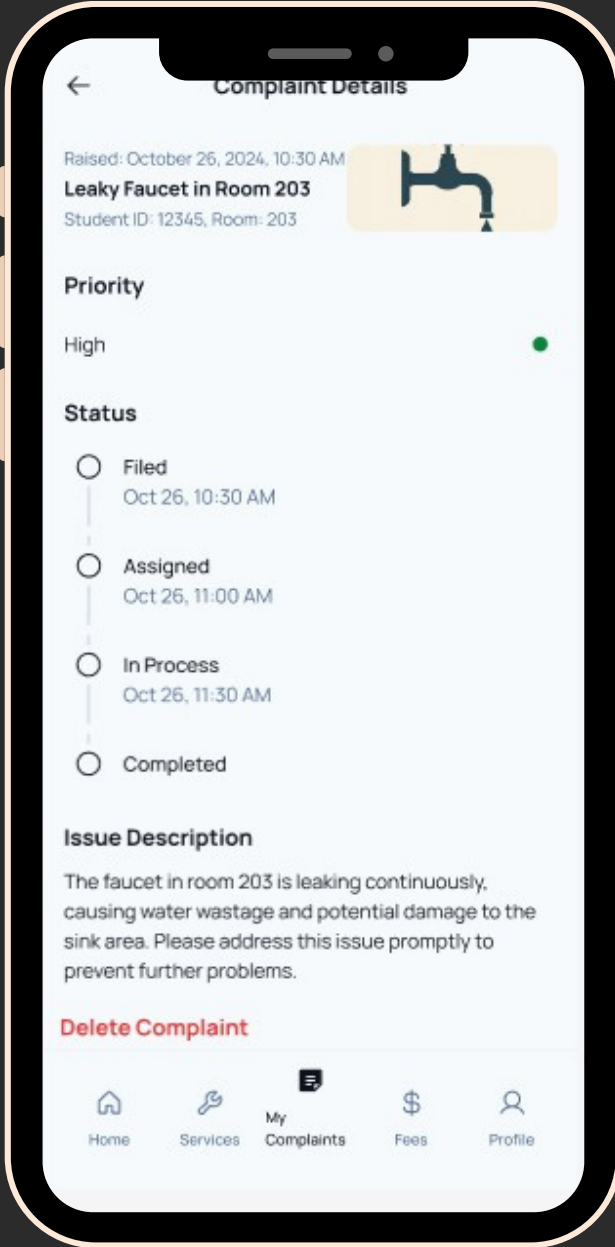
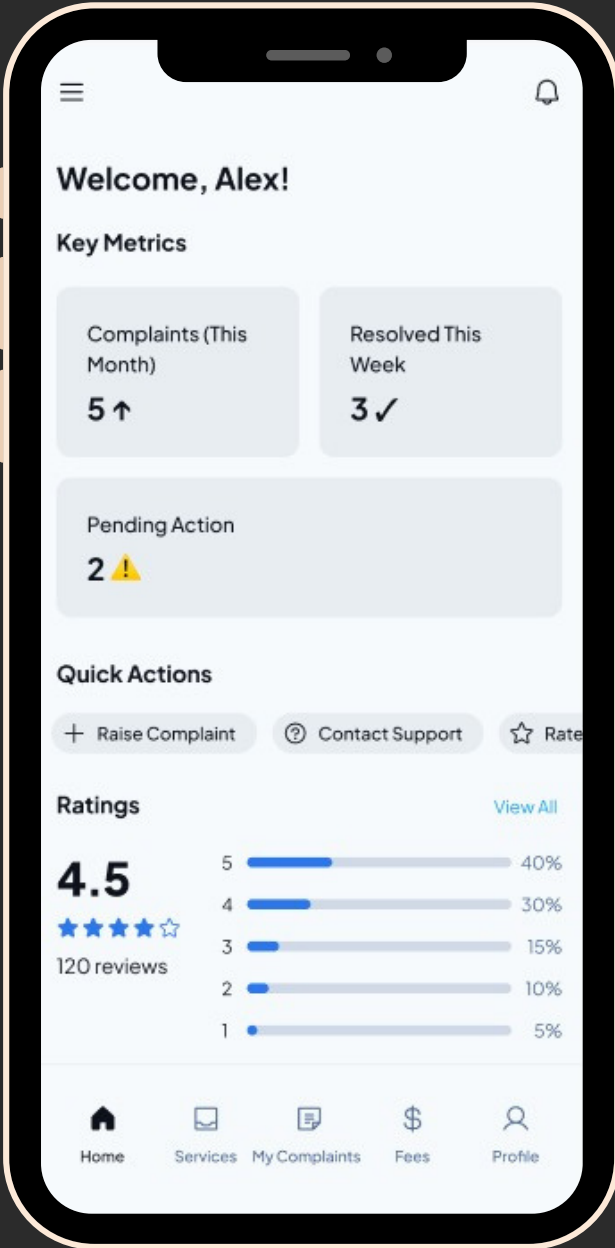
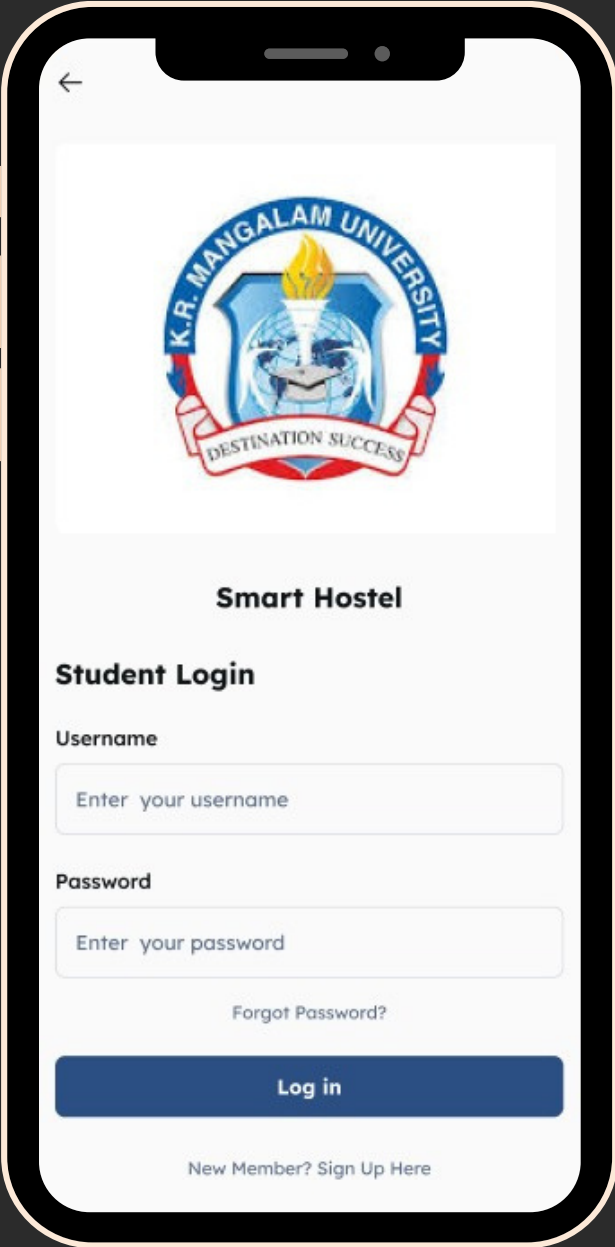
Request ID: B8Sh7i9K5G2KXTlHwxJDeAXiNqYGdwShHRMqET_0M6tUxTQnXbUHdQ==

Ideate: Generating Ideas

7



Prototype



Thank You

2025 November 27